Diabetes Classifier Report

Part 0: Introduction

Background and context: Diabetes is among the most prevalent chronic diseases in the world. Thus, predicting a patient's diabetes status accurately is important to provide patients with an efficient treatment plan on time.

The objective: Determining the best diabetes classifier—which is chosen out of K-Nearest Neighbors (KNN), Decision Tree, and Logistic Regression methods—that can predict the patient's diabetes status based on known factors.

About the dataset: The dataset *diabetes-dataset-2k.csv* consists of eight input variables: 'gender' (Female/Male), 'age' (0–80 years), 'hypertension', 'heart_disease' (0 = No, 1 = Yes), 'smoking_history' ('current', 'ever', 'former', 'never', 'not current', 'No Info'), 'BMI' (10.4-87.7), 'HbA1c_level' (3.5-9.0), and 'blood_glucose_level' (80-300). The response variable, 'diabetes', is binary (0 = No, 1 = Yes).

Steps undertaken: This report first performed Exploratory Data Analysis (EDA) to examine the strength of association between them using box plots (for numerical variables) and contingency tables (for categorical variables). Then, three models KNN, Decision Tree, and Logistic Regression were fitted. After identifying the version with the best goodness-of-fit, I compared the three models using TPR, FPR, precision, ROC curve, and AUC, discussed their pros and cons, and selected the best diabetes classifier.

Part I: EDA: Exploring the variables and association

For the data exploration, 'smoking_history' was regrouped into 'Yes' ('current', 'ever', 'former'), 'No' ('never', 'not current'), and 'No Info'.

a. Categorical variables and 'diabetes'.

Gender and diabetes: There is a difference in diabetes status between females (6.661%) and males (10.943%). The relatively small difference of 4.282 suggests a relatively weak association between gender and diabetes status.

Hypertension and diabetes: 23.333% of people with hypertension have diabetes, compared to 7.297% of those without. The prominent difference of 16.036 suggests a substantial association between hypertension and diabetes status.

Heart disease and diabetes: 36.145% of people with heart disease have diabetes, compared to 7.303% of those without. The pronounced difference of 28.842 suggests a relatively strong association between heart disease and diabetes status.

Smoking history and diabetes: 14.806% of people with a smoking history have diabetes, compared to 9.286% of those without, a difference of 5.52. Meanwhile, 3.745% of people with an unknown smoking history have diabetes, a difference of 5.541. These findings suggest a possible association between smoking history and

diabetes status. However, 721 individuals (roughly 36% of the sample) have unknown smoking histories, meaning there can be uncertainty when determining the strength of this association.

b. Numerical variables and 'diabetes'.

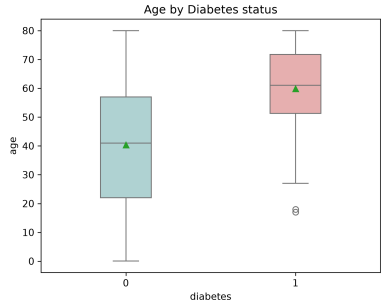


Figure 1. Box plots of age by diabetes status

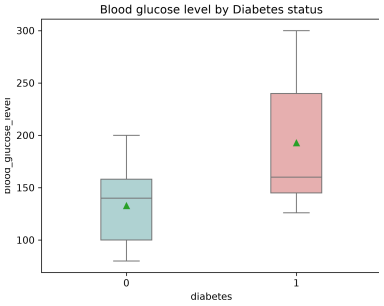


Figure 2. Box plots of Blood glucose level by diabetes status

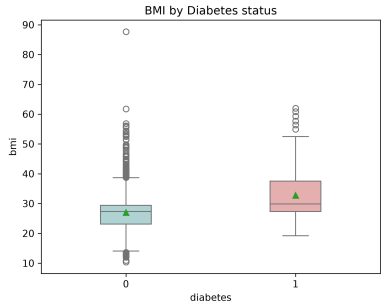


Figure 3. Box plots of BMI by diabetes status

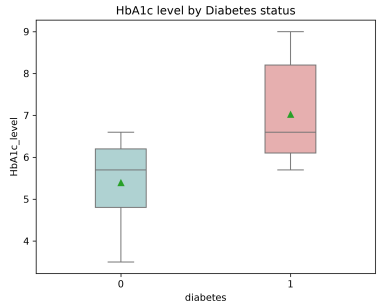


Figure 4. Box plots of HbA1c level by diabetes status

Age and diabetes: Figure 1 shows that the median age of people without diabetes (group 0) is roughly 40, compared to 60 for those with diabetes (group 1). Although the two boxes overlap slightly from 50 to under 60, the difference between their medians suggests a possible association between age and diabetes status.

Blood glucose level and diabetes: Figure 2 shows median blood glucose levels of about 140 and 160 for groups 0 and 1, respectively, with both boxes overlapping from 145 to 160. However, the difference between their means (135 for group 0 and 190 for group 1) is substantial. Additionally, the interquartile range of group 1 is roughly 145-240, compared to 100-160 for group 0. Thus, the middle 50% of individuals with diabetes often have higher blood glucose levels than those without. This suggests a relatively strong association between blood glucose level and diabetes status.

BMI and diabetes: Figure 3 shows group 0's and group 1's means of about 27 and 33, a noticeable difference. However, both groups have close medians at about 27 and 30, and the boxes overlap from roughly 25-30. These observations suggest a relatively weak association between BMI and diabetes status.

HbA1c level and diabetes: Figure 4 shows a small overlap of about 0.1 between the two boxes, with group 0's median of roughly 5.7 and group 1's of 6.5. The difference between the two groups' means is substantial, at about 1.6 (the mean for group 0 is 5.4 and group 1's is 7.0). Moreover, the interquartile range shows that about

75% of people without diabetes have HbA1c levels below 6.2, while about 75% of people with diabetes have HbA1c levels above 6.1. This suggests a relatively strong association between HbA1c level and diabetes status.

Part II: Methods: KNN, Decision Tree and Logistic Regression Classifiers

For data processing, 'gender' was binary encoded into 0 ('Male') and 1 ('Female'). Similarly, 'smoking_history' was binary encoded into 0 ('No', 'No Info') and 1 ('Yes').

For evaluation, I considered the following metrics in descending order of priority: True Positive Rate (TPR), Precision, and False Positive Rate (FPR), where FPR's low priority is because of the dataset and methods' nature. That is, for this report, FPR is more tolerated than False Negative Rate (FNR), or 1 - TPR, as inaccurately labeling patients with diabetes as not having diabetes can postpone medical attention and treatment, which can be fatal.

The dataset was split into a training set and a test set—which comprise 80% and 20% of the original dataset—where the random state was set to 1101. For KNN, the inputs include only quantitative variables; the said input features werethen scaled using **StandardScaler()**. This is because KNN is a distance-based method that is sensitive to differences in feature scales.

a. KNN.

A list of odd integers from 3 to 21 is picked as candidates for K.

Figure 5. Metrics for the top 3-performing versions of the KNN model

K	TPR	FPR	Precision
3	0.559	0.019	0.730769
5	0.529	0.011	0.818182
7	0.471	0.008	0.842105

When I filtered for the KNN with the best TPR, the model where K = 3 is picked as the best version of the KNN method. The candidate value of 3's higher TPR can compensate for its higher FPR and lower precision.

b. Decision Tree.

A list of odd integers 3, 5, 7, 9, 11 is picked as candidate values for max_depth. Similarly, a list of whole numbers 10, 20, 30, 40, 50, 100 is picked as candidate values for min_samples_split. The following is the result of the 4 pairs of depths and splits with the highest TPR, precision, and lowest FPR:

Figure 6. Metrics for the 4 top-performing versions of Decision Tree

max_depth	min_samples_split	TPR	FPR	Precision
11	20	0.706	0.014	0.827586
11	10	0.706	0.011	0.857143
7	40	0.676	0.003	0.958333
7	50	0.676	0.003	0.958333

Note that though max_depth = 11 and min_samples_split = 10 would yield lower FPR and higher precision, max_depth = 11 and min_samples_split = 20 would yield a less complex tree with the same TPR. Hence, the model where max_depth = 11 and min_samples_split = 20 is picked as the best version of the Decision Tree method.

c. Logistic Regression.

The first model, named M1, was generated using **smf.glm**. Then, I determined which input variables to remove based on p-values, as high p-values (above the 0.05 threshold) mean the variable may be insignificant to the model. Examining M1's summary reveals that 'hypertension' and 'smoking_history' both had p-values higher than 0.05. Then, M2, M3, and M4 were built by removing both 'hypertension' and 'smoking_history', only 'hypertension', and only 'smoking_history'.

Filtering the four models by the highest TPR, precision, and the lowest FPR yielded the following table:

Figure 7. Metrics for all versions of the Logistic Regression model

Model	TPR	FPR	Precision
M1	0.559	0.005	0.904762
M2	0.559	0.005	0.904762
M3	0.559	0.005	0.904762
M4	0.559	0.005	0.904762

All models had similar metrics. However, M2 will be chosen because it is less complex than the others due to the removed variables. The following is M2's fitted equation:

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = -26.0722 - 0.6329(\text{gender}) + 1.4376(\text{heart_disease}) + 0.0333(\text{age}) + 0.0982(\text{bmi}) + 2.3065(\text{HbA1c_level}) + 0.0319(\text{blood_glucose_level})$$

For evaluation of the three methods, I also considered the Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC):

Figure 8. Metrics for the three methods

Model	TPR	FPR	Precision	AUC
KNN	0.676	0.007	0.906	0.965312
Decision Tree	0.776	0.010	0.874	0.981400
Logistic Regression	0.635	0.006	0.908	0.955196

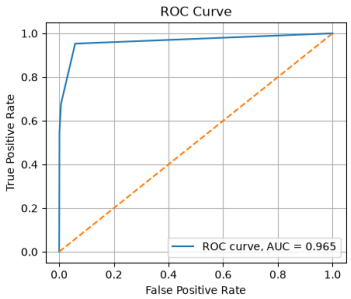


Figure 9. KNN's ROC curve and AUC

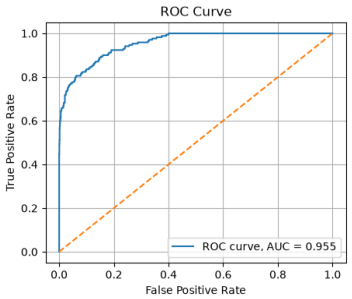


Figure 10. Logistic Regression's ROC curve and AUC

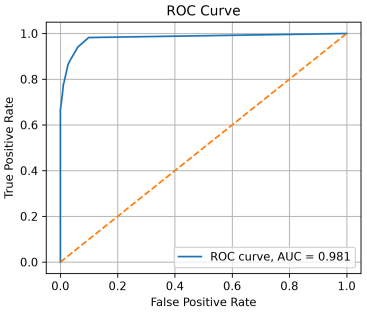


Figure 11. Decision Tree's ROC curve and AUC

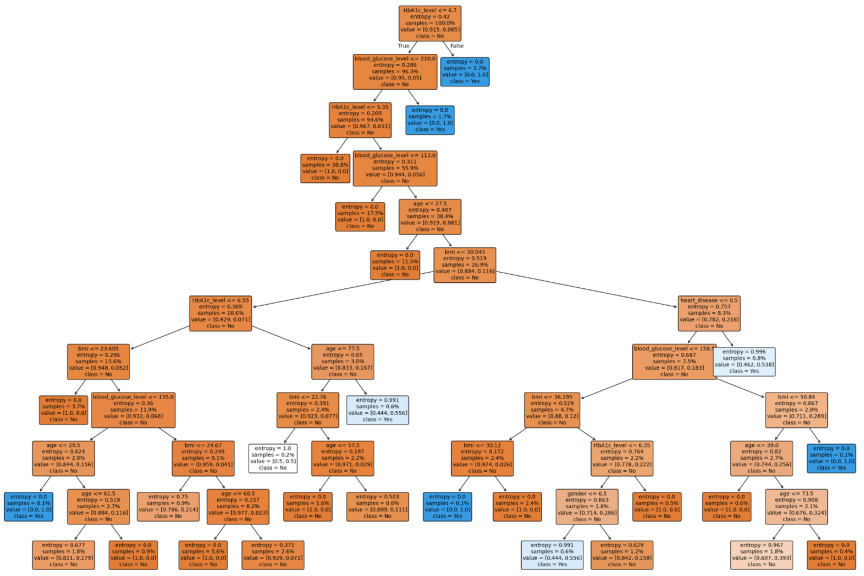


Figure 12. Decision Tree

In Figure 8, when considering the trade-off of TPR and FPR, the Decision Tree model offers the highest TPR that compensates for higher FPR. However, with the max_depth of 11 and the min_samples_split of 20, the Decision Tree model is more complex (Figure 12) and may be prone to overfitting as well as optimistic goodness-of-fit. To examine this, I used the selected models to predict test set test_y, and compared the metrics obtained from this test with those from the full dataset.

Figure 13. Metrics for all three methods when predicting the test set

Model	TPR	FPR	Precision	AUC
Decision Tree	0.706	0.014	0.828	0.933502
Logistic Regression	0.559	0.005	0.905	0.939730
KNN	0.559	0.019	0.731	0.858124

Compared to the first test, the KNN model observed the largest difference, with a reduction of 0.117 in TPR and 0.107 in AUC, followed by Decision Tree (0.070 and 0.048, respectively). Logistic Regression shows the smallest AUC reduction (0.015), suggesting more stability, while the performance for Decision Tree and KNN models may be overestimated.

Part III: Conclusion

Decision Tree offers high TPR yet has potential overfitting due to its complexity. Logistic Regression has stable AUC performance and high precision yet lower TPR and assumes a linear relationship in the log-odds—which may not always be the case. KNN could capture non-linearity yet could not capture categorical variables, and also may be prone to performance overestimation.

Overall, the Decision Tree model has the highest TPR. While its sensitivity to variations in the training set and potential overfitting remain limitations, the Decision Tree's high TPR and accommodation for non-linear relationships make it the best classifier of the three methods.